सिपेट: पेट्रोकेमिकल्स तकनीकी संस्थान (आईपीटी)
रसायन एवं पेट्रोरसायन विभाग
रसायन एवं उर्वरक मंत्रालय, भारत सरकार
गिण्डी, चेन्नई - 600 032.
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ई-मेल : chennai@cipet.gov.in
वेब्सइट : www.cipet.gov.in

सिपेट CIPET

ITUWTSA NEW DELHI 2024

CIPET: INSTITUTE OF PETROCHEMICALS TECHNOLOGY (IPT)
Department of Chemicals & Petrochemicals
Ministry of Chemicals & Fertilizers, Govt. of India
Guindy, Chennai - 600 032.
Phone : 91-44-2225 4701 (6 Lines)
E-mail : chennai@cipet.gov.in
Website : www.cipet.gov.in

संदर्भ/Ref: परीक्षण/TR/ 25081795 दिनांक/Date: 30-03-2026

प्रति /To,

ADITYA POLYMERS

Plot No. SP-39, (Phase-1), PIPDIC Industrial Estate, Sedarapet, Villiyanur Commune, Puducherry - 605 111

विषय/Sub : Testing of samples-reg.

संदर्भ/Ref : Your letter no. --

Dated: 19-08-2025

महोदय,

कृपया हमारी परीक्षण रिपोर्ट क्रमांक प्राप्त करे। Please find enclosed herewith our Test Report no.	84501	दिनांक dated	30-03-2026

धन्यवाद,

भवदीय

T. Bhu

अधिकृत हस्ताक्षरकर्ता

संलग्न : उपर्युक्तानुसार
Encl:1.a/a.

केंद्रीय पेट्रोरसायन अभियांत्रिकी एवं प्रौद्योगिकी संस्थान
पेट्रोकेमिकल्स तकनीकी संस्थान
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परीक्षण रिपोर्ट / TEST REPORT

को जारी / Issued to :
ADITYA POLYMERS
Plot No. SP-39, (Phase-1), PIPDIC Industrial Estate,
Sedarapet, Villiyanur Commune, Puducherry - 605 111

क्र.सं / **Sl. No.** 38504

रिपोर्ट सं / **REPORT NO. : 84051**

दिनाक / Date : **30-03-2026**

Pages............Nos. Part A,B,C & D

--

संदर्भ / **Customer Let. Ref :** 19-08-2025

परीक्षण मानक स्तर के अनुसार परीक्षण रिपोर्ट / **TEST REPORT AS PER TEST STANDARD : Refer Part C**

भाग - क / PART - A

प्रस्तुत सैंपिल का विवरण / PARTICULARS OF SAMPLE SUBMITTED

अ)	सैंपिल का नाम / a) Name of the Sample	: Compostable Film - as stated by party
आ)	सैंपिल प्राप्त होने की तारीख / b) Date of Receipt of sample	: 20-08-2025
इ)	ग्रेड/प्रकार/आकार/वर्ग / c) Grade / variety / type / size / class	: Nil
ई)	घोषित मूल्य / d) Declared value, If any	: Nil
उ)	कोड सं. / e) Code No.	: Nil
ऊ)	बैच सं. एवं निर्माण तारीख / f) Batch No. and Date of Manufacture	: Nil
ऋ)	मात्रा / g) Quantity	: 2 kg
ए)	पेंकिंग की रीति / h) Mode of Packing	: Packed in envelope
ऐ)	मोहर बंद या नहीं / i) Sealed or not	: Sealed
ओ)	कोई अन्य सूचना / j) Any other information	:

25081795

भाग - ख / PART - B

अनुपूरक सूचनाएँ / SUPPLEMENTARY INFORMATIONS

अ)	सैंपिलिंग कार्यवाहियों हेतु संदर्भ / a) Reference to sampling procedure	: Sampling not done by this lab
आ) b)	माप करने हेतं लिए गए सहायक दस्तावेज एवं प्राप्त परिणाम Supporting documents for the measurement taken and result derived	: As given in Part C
इ) c)	संबंधित कार्य अनुदेशों में निर्धारित के अनुसार परीक्षण रीति से कोई परिवर्तन Deviation from the test method as prescribed in relevant work instructions, if any	: No deviation from the standard

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परीक्षण रिपोर्ट / TEST REPORT

रिपोर्ट सं / **REPORT NO. :84051**

क्र.सं / Sl. No. 38504

दिनाक / Date : **30-03-2026**

भाग - ग / PART - C

परीक्षण परिणाम / TEST RESULTS

Test Duration : 28-08-2025 to 30-03-2026

sl. No.	Name of the test	Test method / Standard	unit	Results Obtained	Specific Requirements
1	Material Identification	FTIR & DSC	-	Blend of Polylacticacid and Polybutyleneadipate-co-terephthalate (PLA & PBAT)	-
2	Disintegration (Dry mass remains in 2 mm sieve after 84 days)	ISO 17088:2021 Cl.no. 6.2	%	8.3	Not more than 10
3	Ultimate aerobic Biodegradation (with reference to 100% degradation of positive reference)	ISO 17088:2021 Cl.no. 6.3	%	91.1 (At the end of 140 days)	> 90 (At the end of the test period not more than 180 days.)
4	Plant growth test				
	Monocotyledon (Onion) % Seed emergence	OECD 208 Guidelines	%	96	> 90
	Dicotyledon (Fenugreek) % Seed Emergence		%	94	> 90
5	Acute Ecotoxic effects to earthworm	ISO 11268-1 : 2012	%		
a	Survial of adult earthworm at the end of 7 days			100	Shall be more than 90
b	Survial of adult earthworm at the end of 14 days			98	Shall be more than 90
c	Biomass end of the 14 days			95	Shall be more than 90
6	Chronic ecotoxic effets to earthworm	ISO 11268-2 : 2023	%		
a	Survial of adult earthworm at the end of 28 days			96	Shall be more than 90
b	Survial of adult earthworm at the end of 56 days			97	Shall be more than 90
c	Offspring at the end of 56 days			96	Shall be more than 90
d	Biomass end of the 56 days			97	Shall be more than 90

The detailed observation on biodegradability test is enclosed as Annexure

AUTHORISED SIGNATORY
Dr. Radhashyam Giri

AUTHORISED SIGNATORY
T.Bharani
Technical Officer

रिपोर्ट सं / REPORT NO. : 84051 | परीक्षण परिणाम / TEST RESULTS | दिनाक /Date : 30-03-2026

Issued to: Aditya Polymers, Plot No. SP-39, (Phase-1), PIPDIC Industrial Estate,
Sedarapet, Villiyanur, Commune, Puducherry - 605 111

Customer Letter No. Nil, dtd. 11/08/2025

PART C - TEST RESULTS

Sl.No	Name of the Test	Test Method/ Standard	Unit	Results Obtained	Specified Requirements*
7	Heavy metals concentration				
a.	Arsenic (As)	ISO 17088:2021	mg/kg	Not Detected	10
b.	Copper (Cu)			0.0401	300
c.	Nickel (Ni)			0.057	50
d.	Zinc (Zn)			0.0221	1000
e	Cobalt (Co)			Not Detected	-
f.	Chromium (Cr)			0.3349	50
g.	Molybdenum (Mo)			0.3573	-
h.	Mercury (Hg)			Not Detected	0.15
i.	Cadmium (Cd)			0.0035	5
j.	Lead (Pb)			0.0857	100
k.	Selenium (Se)			0.1716	-

* Based on Municipal solid waste (Management and Handling) Rules, 2016 notified on 8th April, 2016 by Ministry of Environment , Forests and climate change, Government of India. Note that concentration of metals like cobalt, molybdenum, and selenium is not mentioned in the notification.

NIL

Note

1. This Test Report / Certificate is issued only for the samples submitted to the laboratory.
2. The results stated above related only to the items tested.
3. The quality of the subsequent production lot has to be ensured by the purchaser.
4. This Test Report shall not be reproduced except in full without the written approval of the laboratory.
5. Any anomaly/discrepancy in this report should be brought to the notice of the laboratory within 30 days from the date of issue.
6. Subcontracted Tests (if any): Nil

AUTHORISED SIGNATORY
Dr. Radhashyam Giri
Technical Officer

AUTHORISED SIGNATORY
T.Bharani

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Customer Letter No. Nil, dtd. 11/08/2025

OBSERVATION FOR BIODEGRADABILITY TEST AS PER ISO 17088:2021

1 **Sample Details (As stated by Party):** Compostable Film

2 **Material Identification by FTIR** : Blend of Polylacticacid and Polybutyleneadipate-co-terephthalate (PLA & PBAT)

BIODEGRADABILITY TEST AS PER ISO:14855-1

3 **Observation**

(i) Conditions of reaction mixtures

Origin of Compost: Livestock excrement, municipal and vegetable waste

Reaction Temperature (^{o}C)	:	58
Dry Solid (%)	:	53.42
Volatile content (%)	:	17.02
CO_2 evolved during first 10days in blank	:	50.64 mg/g
Test duration (days)	:	140 days
Reference material	:	Cellulose
Volume of reaction vessel (mL)	:	3000 ml

(ii) pH of test medium

S.No.	Compost Vessel	pH (Before)	pH (After)
1	Blank 1	7.34	7.32
2	Blank 2	7.35	7.33
3	Blank 3	7.31	7.28
4	Cellulose 1	7.19	7.16
5	Cellulose 2	7.23	7.21
6	Cellulose 3	7.21	7.18
7	Negative 1	7.16	7.15
8	Negative 2	7.12	7.1
9	Negative 3	7.14	7.12
10	Sample 1	7.11	7.08
11	Sample 2	7.13	7.12
12	Sample 3	7.17	7.15

Contd.,

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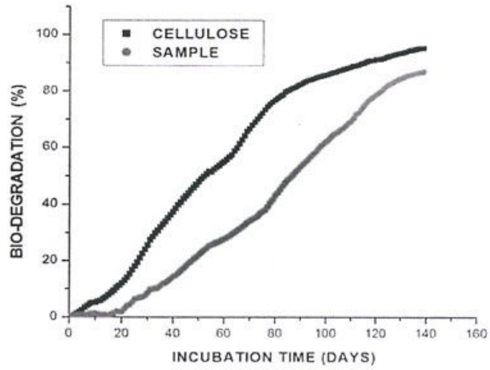
4 **Result:** Percentage biodegradation relative to positive reference

Sample (Mean) : 91.1 % at the end of 140 days

The reference Material - cellulose : ~ 100%



5 **Visual Observation of Sample**

Description	Week 1-4	Week 5-8	Week 9-12
Structure	Cut pieces	Cut pieces	Fragmented pieces
Moisture	Adequate moisture Level	Adequate moisture Level	Adequate moisture Level
Colour	White	Dirty	Dirty
Fungal Development	Nil	Nil	Nil
Smell	Organic/dirt like	Organic/dirt like	Organic/dirt like

Description	Week 13-16	Week 17-18	Week 19-20
Structure	Fragmented pieces	Fragmented pieces	Fragmented pieces
Moisture	Adequate moisture Level	Adequate moisture Level	Adequate moisture Level
Colour	Dirty	Dirty	Dirty
Fungal Development	Nil	Nil	Nil
Smell	Organic/dirt like	Organic/dirt like	Organic/dirt like

Contd.,

AUTHORISED SIGNATORY
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CIPET : IPT - Chennai -32

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Continuation Sheet

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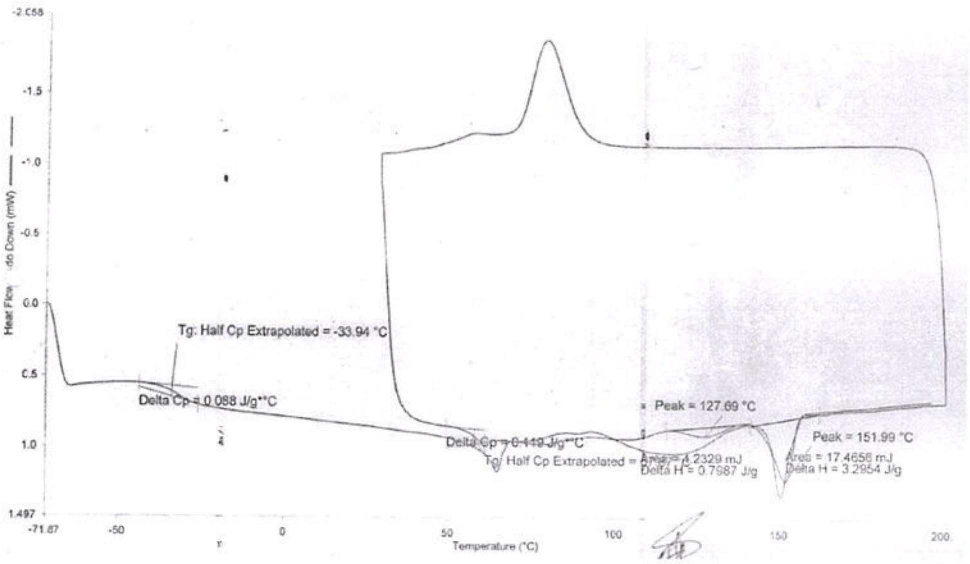
6 **Visual Observation of Compost**

Description	Week 1-4	Week 5-8	Week 9-12
Structure	Fine Particles	Fine Particles	Fine Particles
Moisture	Adequate moisture Level	Adequate moisture Level	Adequate moisture Level
Colour	Dark Brown	Dark Brown	Dark Brown
Fungal Development	Nil	Nil	Nil
Smell	Organic/dirt like	Organic/dirt like	Organic/dirt like

Description	Week 13-16	Week 17-18	Week 19-20
Structure	Fine Particles	Fine Particles	Fine Particles
Moisture	Adequate moisture Level	Adequate moisture Level	Adequate moisture Level
Colour	Dark Brown	Dark Brown	Dark Brown
Fungal Development	Nil	Nil	Nil
Smell	Organic/dirt like	Organic/dirt like	Organic/dirt like

7 **DSC Analysis**



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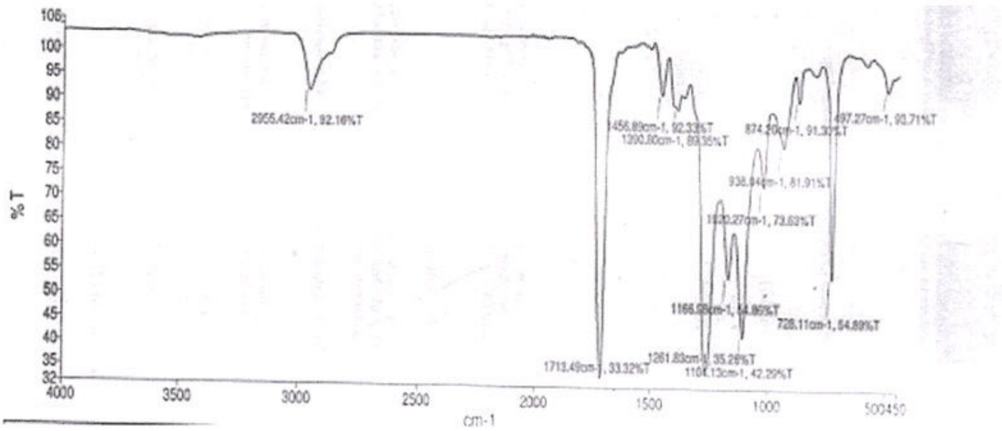
8 **FTIR Analysis**



FTIR Interpretation

Wave number (cm^{-1})	Nature of Bond
2955	O-H stretch
1713	C=O stretch in PLA and PBAT
1456	$-CH_2$ Plane Bending
1261	C-O Stretch
1104	C-O bend of PBAT
1020	C-O bend of PBAT
874	$O-CH-CH_2-H$ of ester
728	CH_2 Plane of benzene ring

Comment: The above DSC & FTIR analysis indicates the above sample is Blend of Polylacticacid and Polybutyleneadipate-co-terephthalate (PLA & PBAT)

Contd.,

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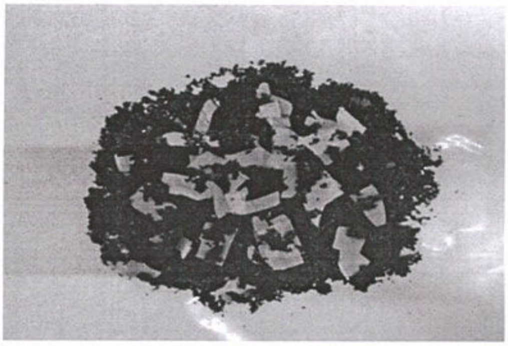
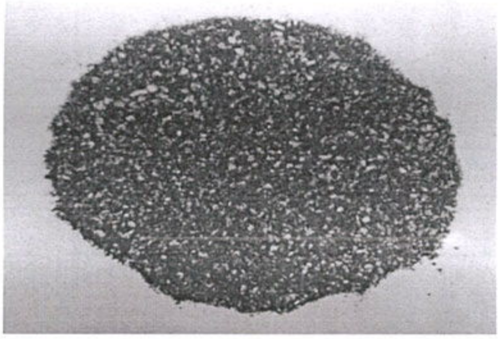


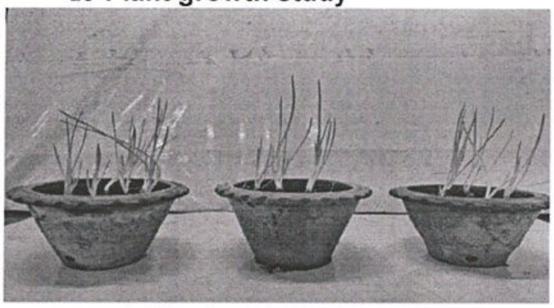
Before Disintegration



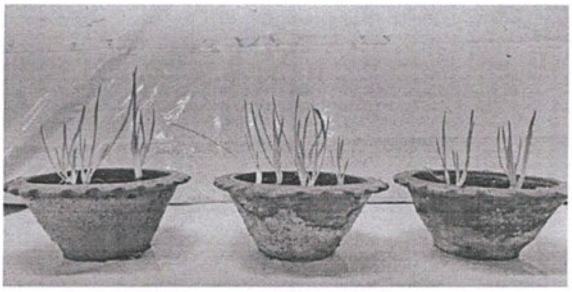
After Disintegration

The disintegration of the supplied sample by passing through 2 mm sieve after 12 week in composting condition as per ISO 17088-2021 was found not more than 10% of original dry mass remain.

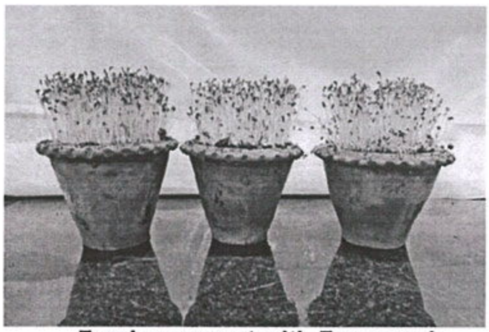
10 **Plant growth study**



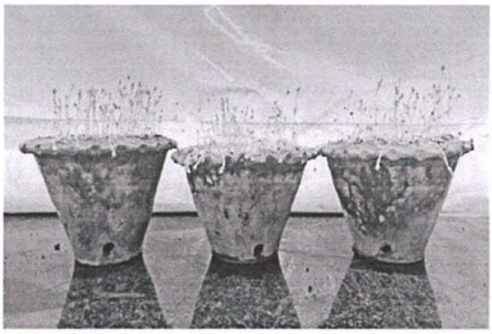
Fresh compost with Onion



Bio-degradation tested compost with Onion



Fresh compost with Fenugreek



Bio-degradation tested compost with Fenugreek

The percentage of seed germination rate is found to be greater than 90% for both Onion and Fenugreek .

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